Perception Driven Robot Grasping

Advised By: Jikai Wang, Qifan Zhang

Team: Eugene Wang, Kevin Cui, Lucas Wu, Edgar Poon, Pai Zheng, Ellie Tang, Ted Ma

DR. YU XIANG | UTD – 013 | CAST STEM Bridge

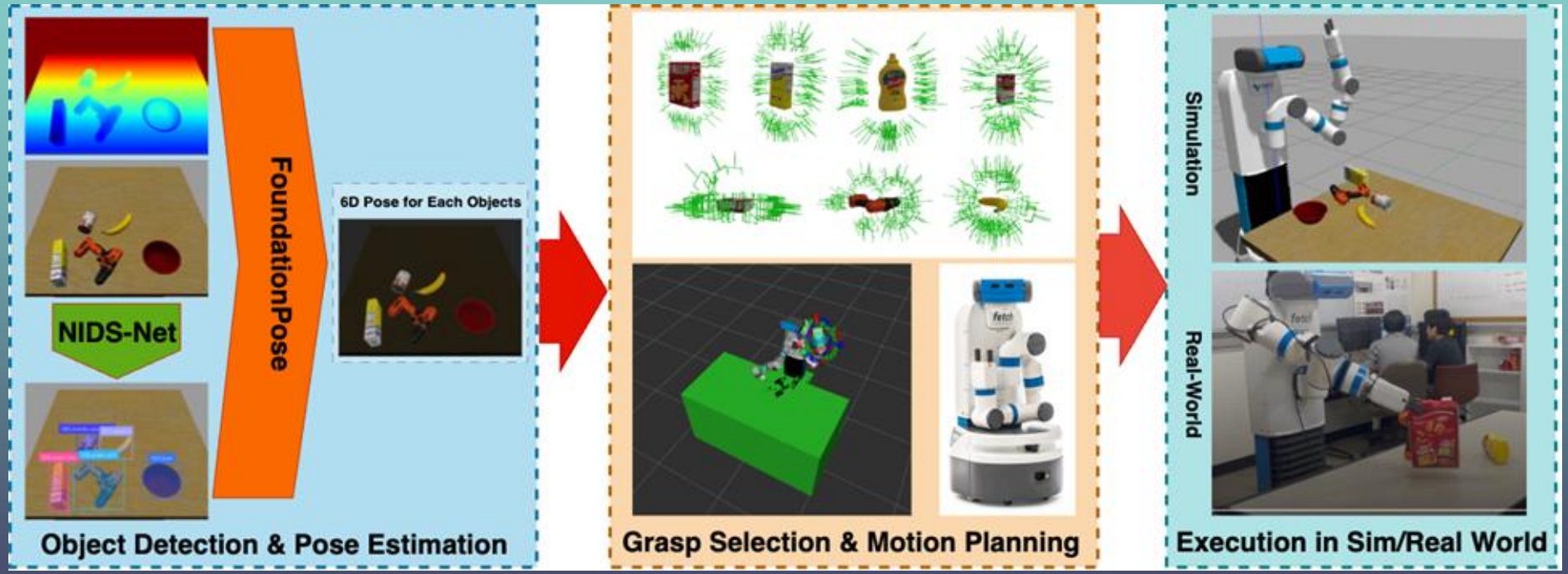
Project Description

Grasping diverse objects in unstructured environments requires integration of object recognition & motion planning.

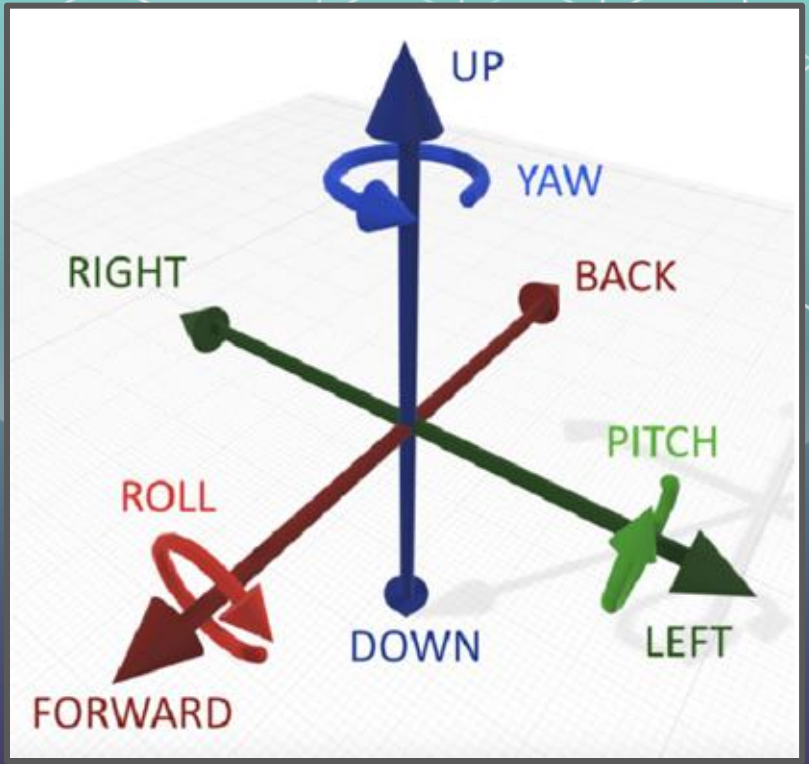
Over the past 6 weeks, we developed a multimodal pipeline for grasp planning in simulation & real world.

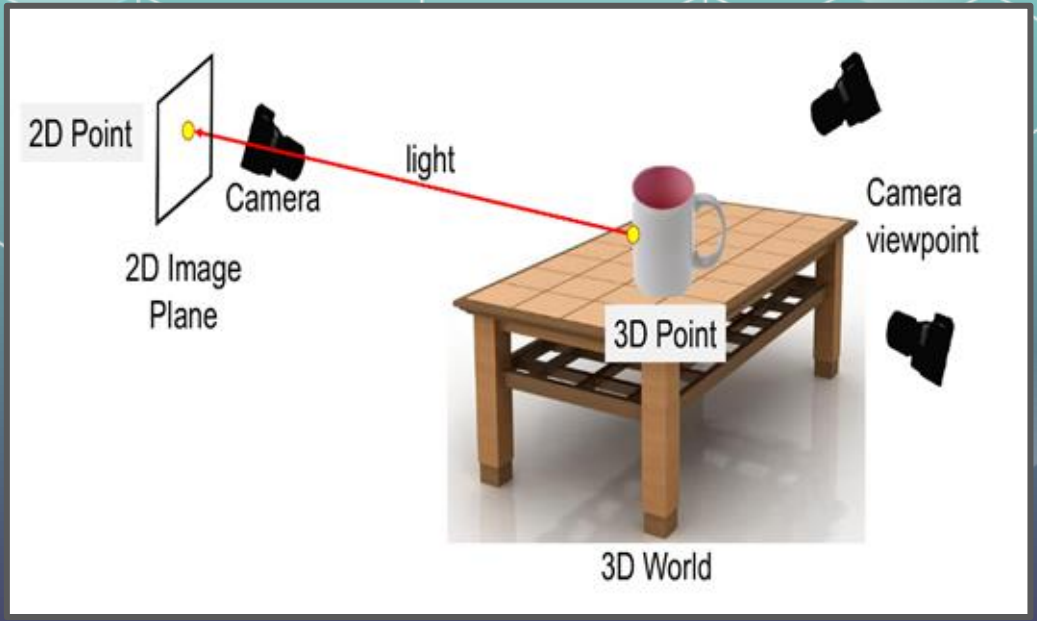
- Detection & Segmentation: NIDS-Net
- 6D Pose Estimation: FoundationPose
- Grasping: MoveIt + YCB objects
- Setup: Gazebo + Fetch robot

Our Project Pipeline



Computer Vision Basics: Part 1





Computer Vision Basics Part 2

	2D Transformations	3D Transformations
Translation	$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$ augmented vector $\bar{\mathbf{x}} = (x, y, 1)$ $\bar{\mathbf{x}}' = \underbrace{\begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}}_{3 \times 3} \bar{\mathbf{x}}$ Degree of Freedom: 2 (tx, ty)	$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$ augmented vector $\bar{\mathbf{x}} = (x, y, z, 1)$ $\mathbf{x}' = \underbrace{\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}}_{3 \times 4} \bar{\mathbf{x}}$ Degree of Freedom: 3 (tx, ty, tz)
Euclidean Transformation (rotation + translation)	$\mathbf{x}' = \mathbf{R}\mathbf{x} + \mathbf{t}$ $\bar{\mathbf{x}} = (x, y, 1)$ $\mathbf{x}' = \underbrace{\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}}_{2 \times 3} \bar{\mathbf{x}}$ $\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ Degree of Freedom: 3 (tx, ty, θ)	$\mathbf{x}' = R\mathbf{x} + \mathbf{t}$ $\bar{\mathbf{x}} = (x, y, z, 1)$ $\mathbf{x}' = \underbrace{[R \mid \mathbf{t}]}_{3 \times 4} \bar{\mathbf{x}}$ $R_{3\times3} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix}$ Degree of Freedom: 6 (tx, ty, tz, θxy, θxz, θyz)

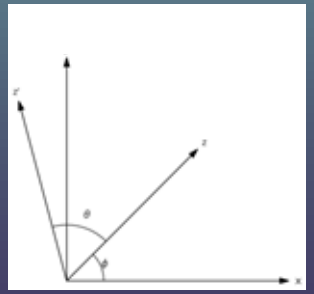
Hierarchy of 2D Transformations

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}_{2\times3}$	2	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}_{2\times3}$	3	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix}_{2\times3}$	4	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{2\times3}$	6	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{3\times3}$	8	straight lines	

3D Transformations

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}_{3\times4}$	3	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}_{3\times4}$	6	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix}_{3\times4}$	7	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{3\times4}$	12	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{4\times4}$	15	straight lines	

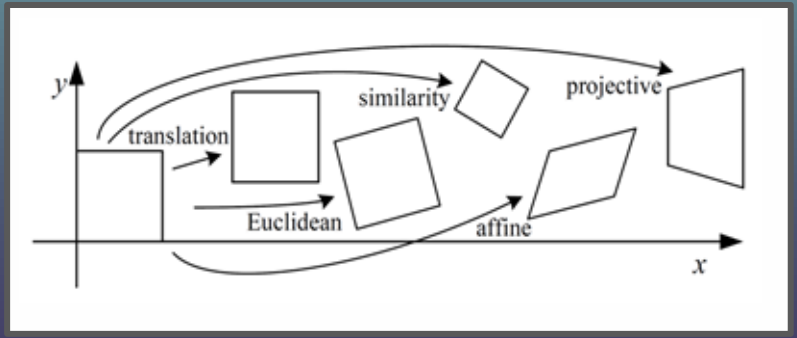
Rotation of a Complex Number



$$z = pe^{j\phi} = p\cos\phi + j\,p\sin\phi = x + jy,$$

$$\begin{aligned} z' = pe^{j(\phi+\theta)} &= p\cos(\phi+\theta) + j\,p\sin(\phi+\theta) \\ &= p(\cos\phi\cos\theta - \sin\phi\sin\theta) + j\,p(\sin\phi\cos\theta + \cos\phi\sin\theta) \\ &= (x\cos\theta - y\sin\theta) + j\,(y\cos\theta + x\sin\theta) = x' + jy'. \end{aligned}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

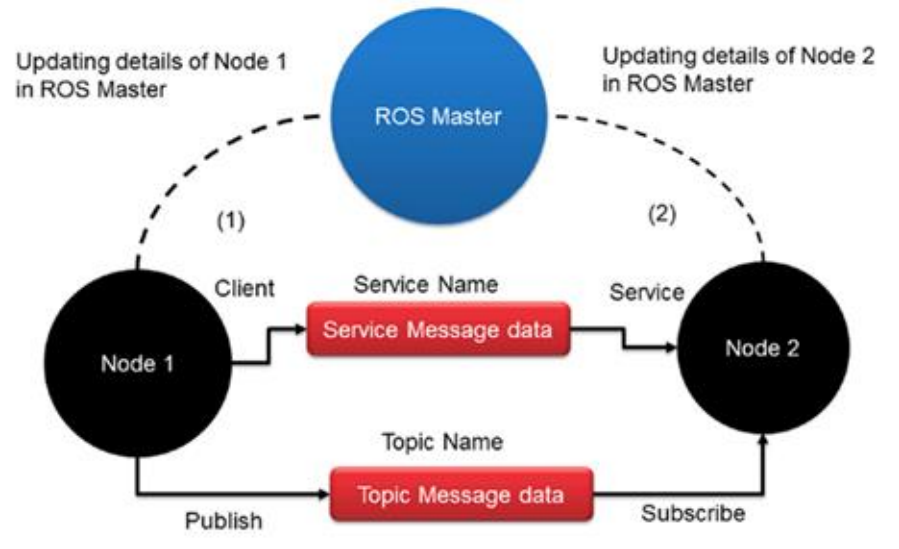


Robot Operating Systems (ROS)

The ROS enables communication between software and robot hardware.

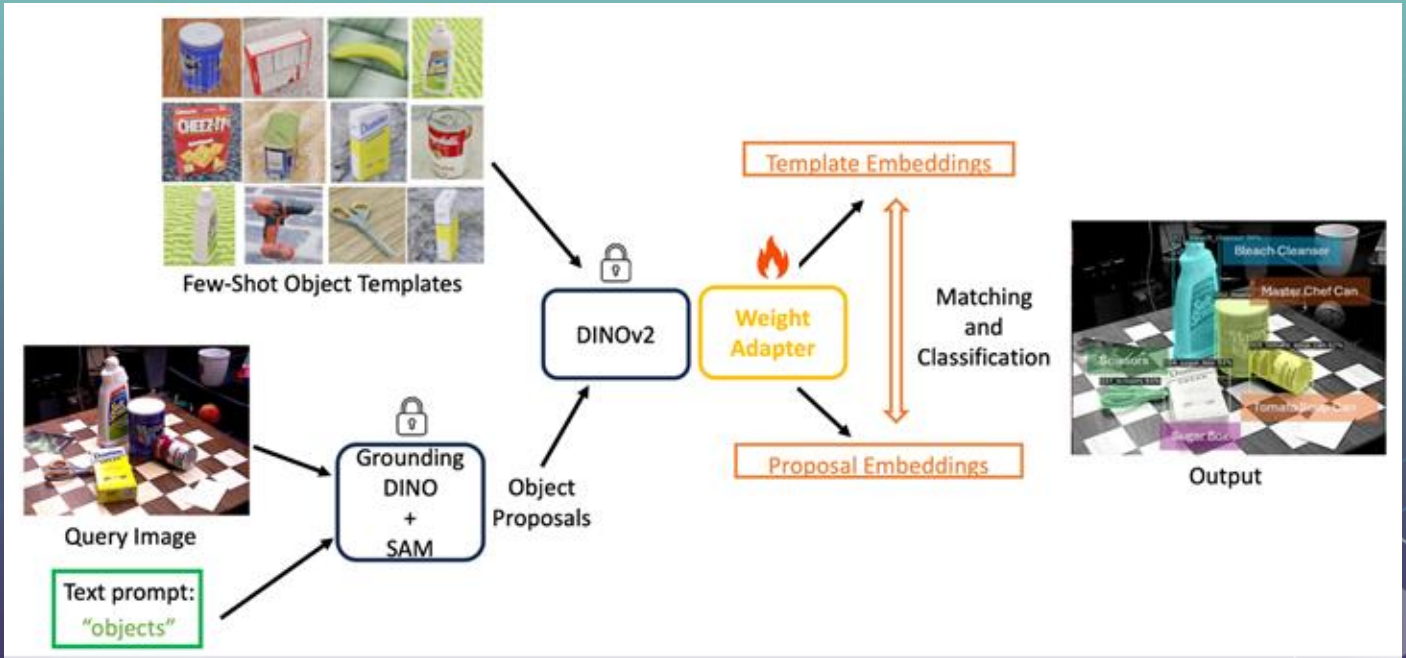
Main ROS Components

- **Nodes**: perform specific tasks (e.g., control gripper, read sensors)
- **Topics**: communication channels between nodes
- **Messages**: data sent over topics
- **Services**: one-time requests (e.g., open gripper)



Object Detection & Segmentation: NIDS-Net

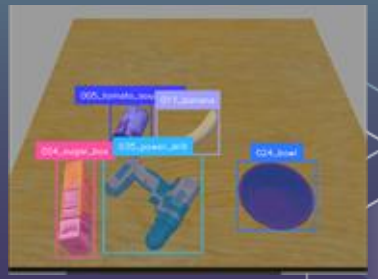
NIDS-Net: Novel Instance Detection and Segmentation (NIDS) given a few examples of each instance.



NIDS-Net: Practice

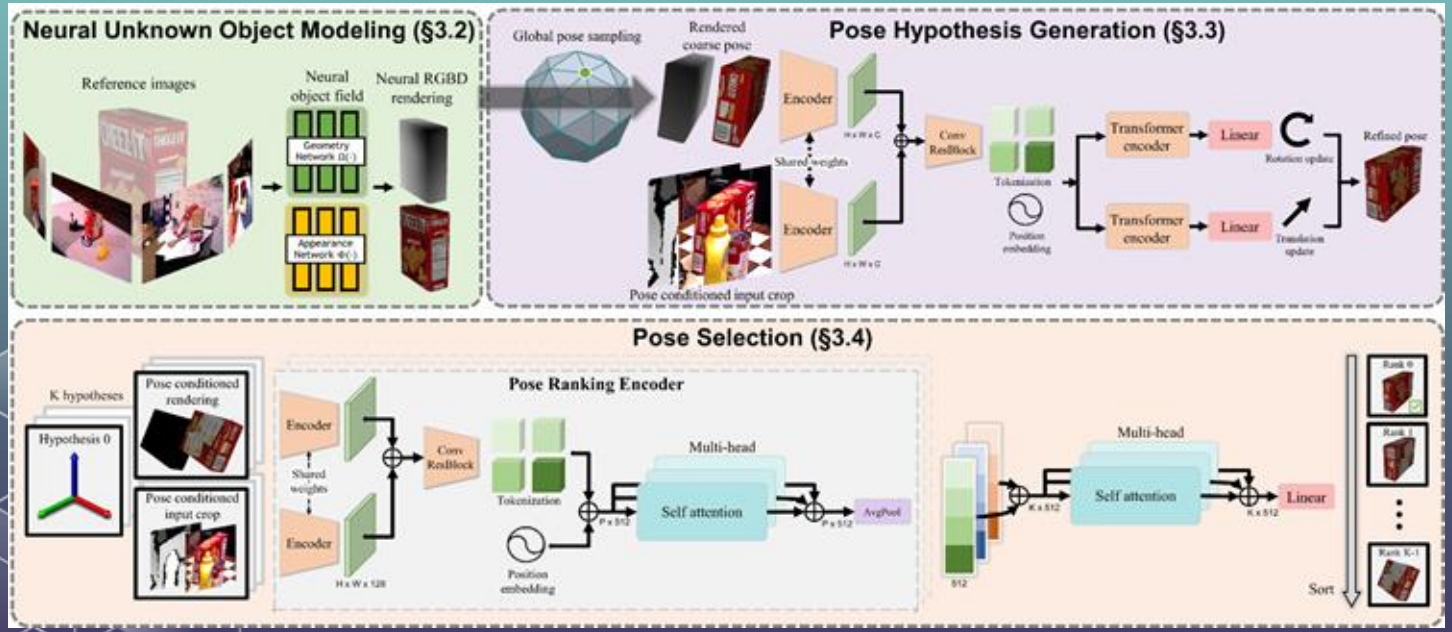
1. **Inputs:** RGB images taken from the **Fetch Gazebo simulation and real-world**
2. **Outputs: the NIDS-Net** to generate the segmentation masks and detection labels.

Template images



Object Pose Estimation: FoundationPose

FoundationPose is a <u>unified</u> foundation model for 6D object pose estimation and tracking.

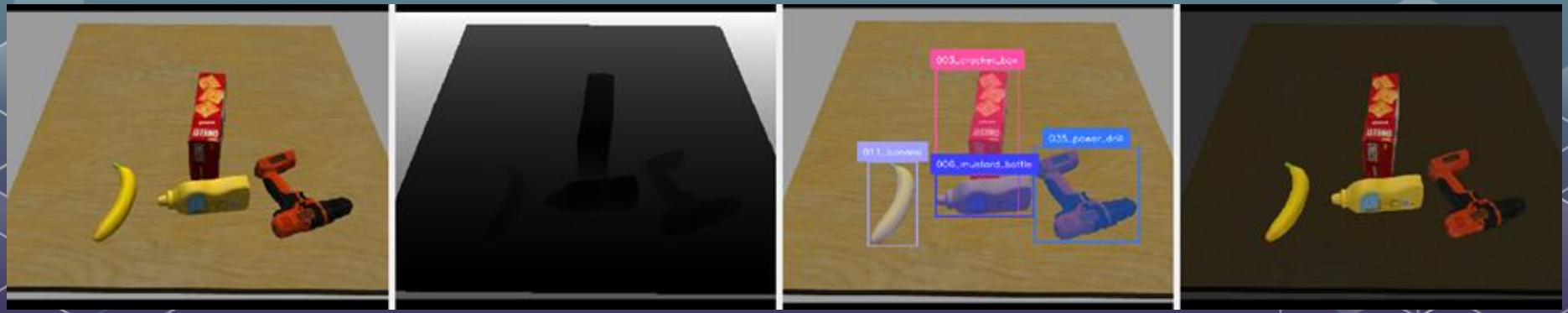


- Trained on <u>Large-scale</u> Synthetic Dataset
- <u>Transformer-based Architecture</u>

FoundationPose: Practice

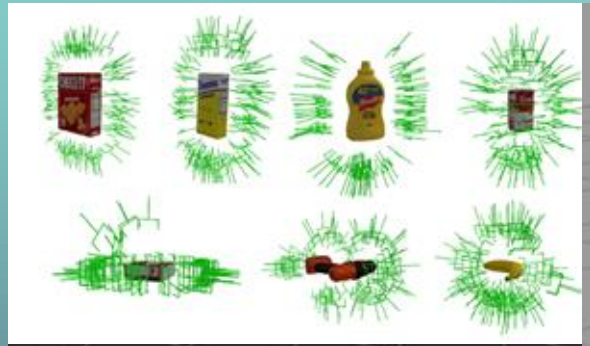
1. **Inputs:**
    a. RGB & Depth Images from the Robot Camera
    b. Segmentations from NIDS-Net outputs
2. **Outputs:**
    a. The estimated 6DoF pose in camera space
3. **The Object Poses will be passed to MoveIt for Motion Planning.**

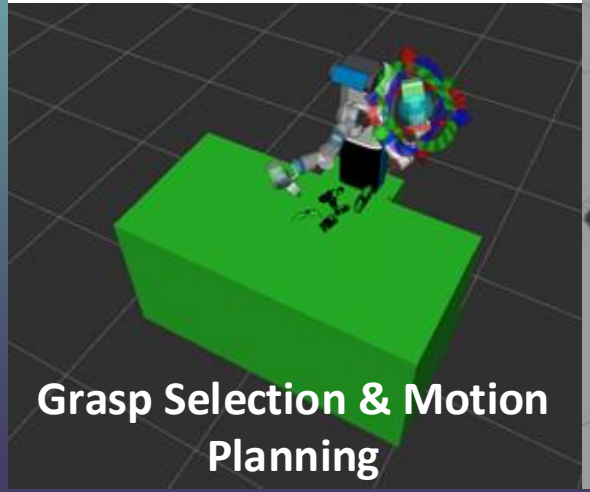
RGB Image **Depth Image** **Segmentation (NIDS-Net)** **Rendered Pose**



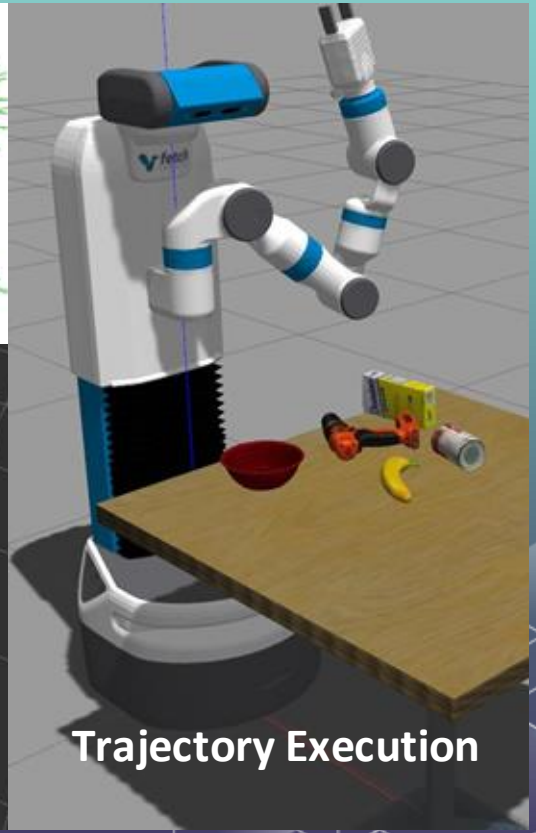
MoveIt: Motion Planning

- Motion Planning
- Collision Avoidance
- Kinematics Solving
- Trajectory Execution
- Integration with Perception





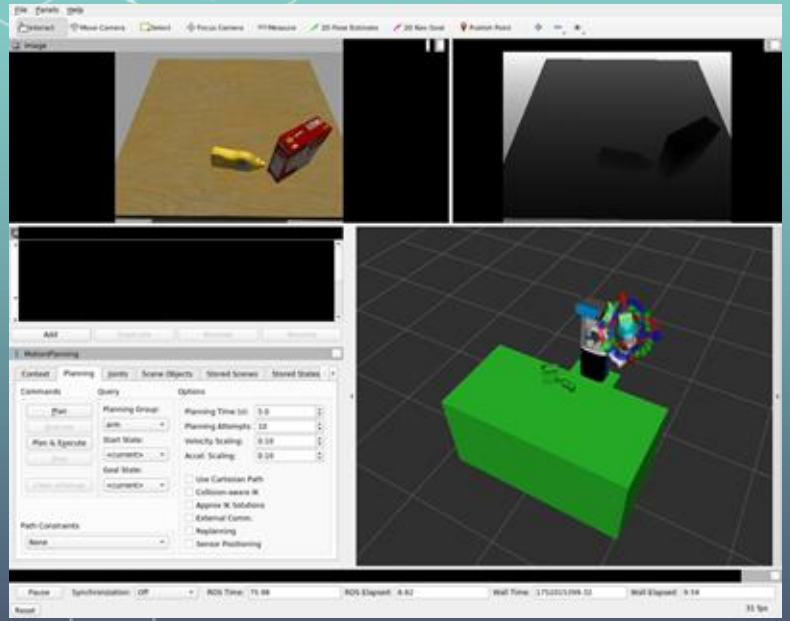
Grasp Selection & Motion Planning



Trajectory Execution

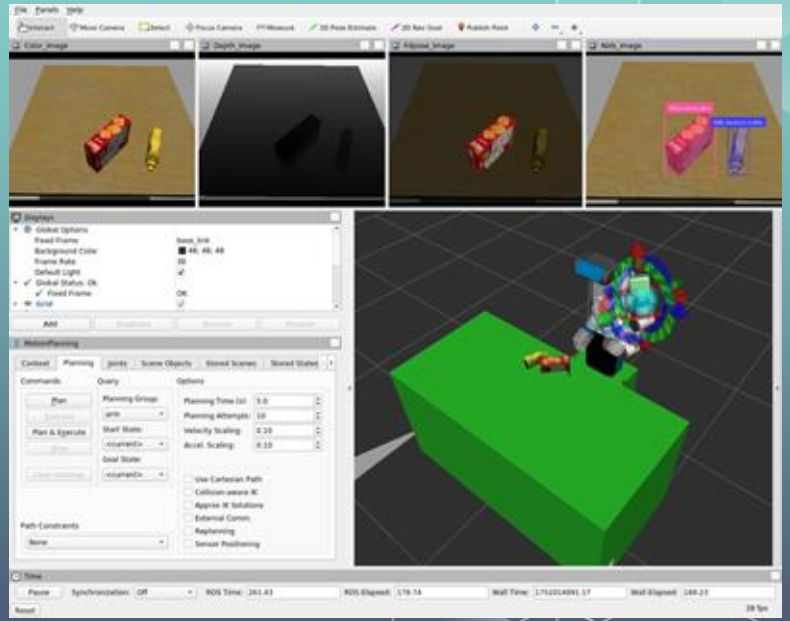
Verification in the Simulation

Groundtruth Pose



Directly get the pose from Gazebo

Estimated Pose



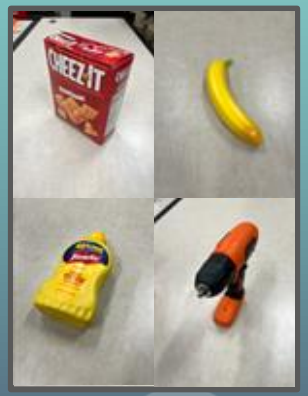
Use the NIDS-Net to do the segmentation and get the estimated pose through Foundation Pose

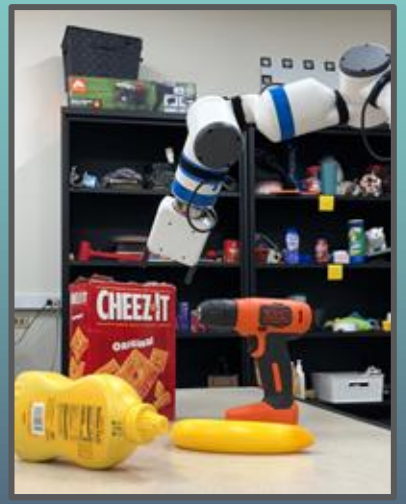
Running Grasping on a Real Fetch Robot

Step 1: Experiment Setup

Step 2: Grasping on Fetch Robot

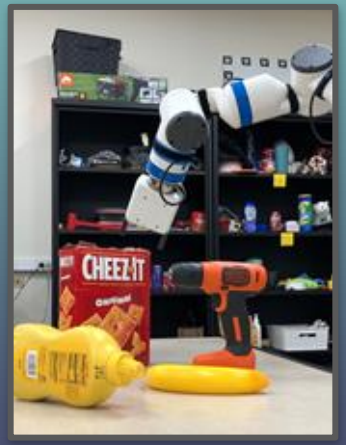
01: Experiment Setup



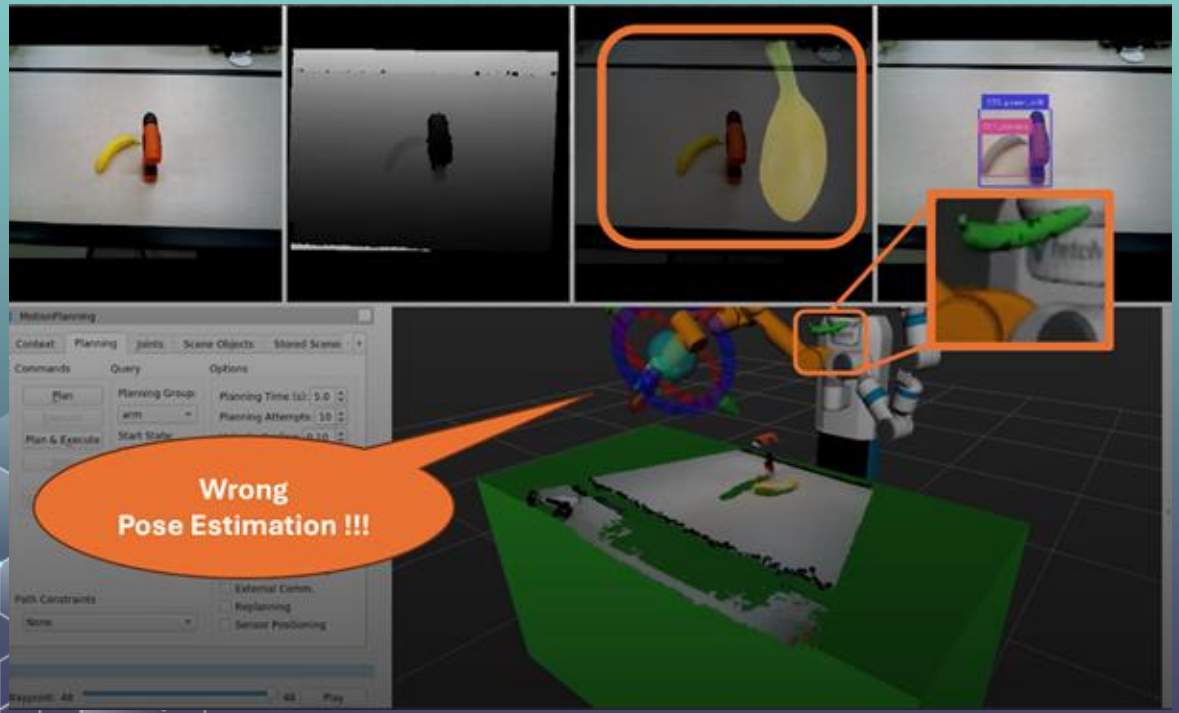


02: Grasping on Fetch Robot

Once a FULL grasp is found, the Fetch robot will execute the planned grasping motion.

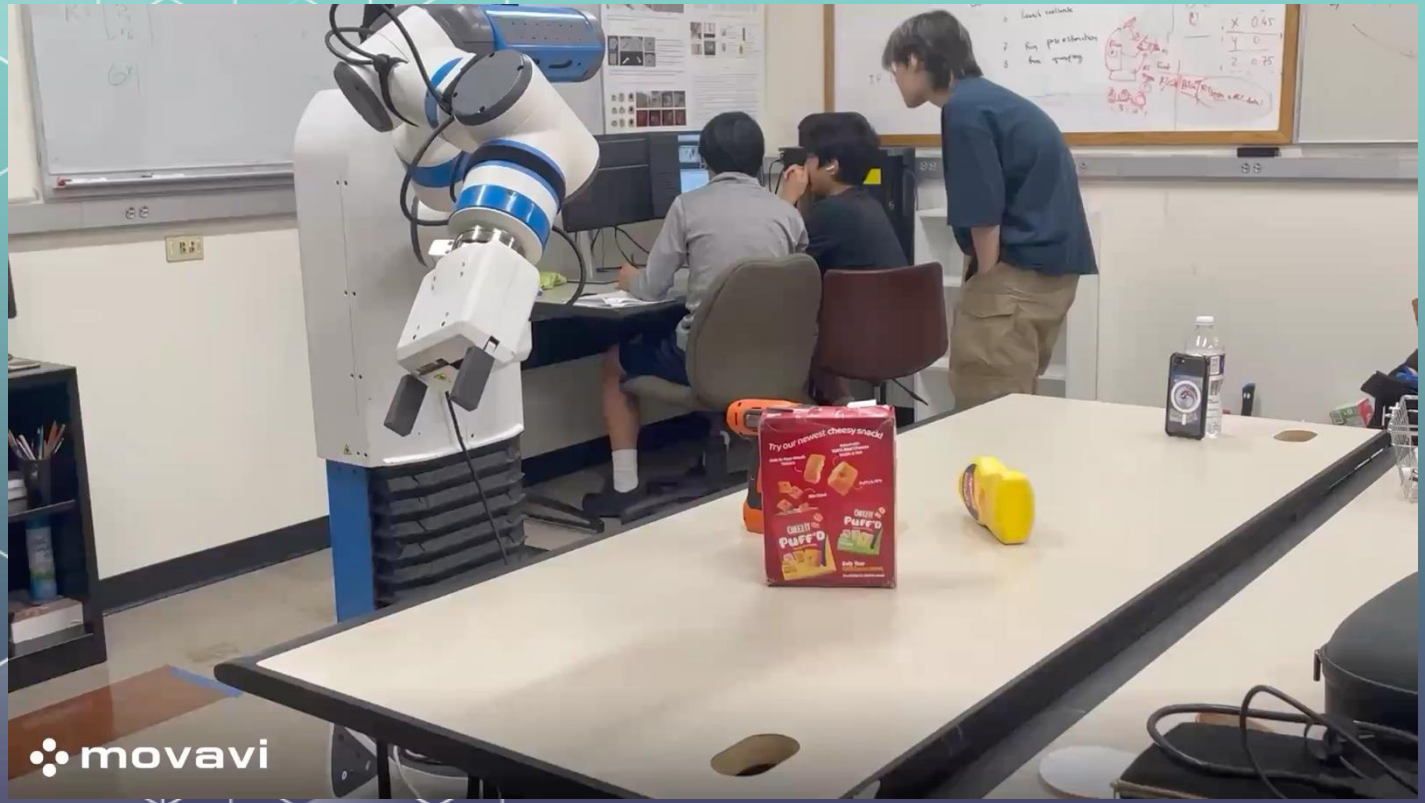


Limitations and Failure Cases



- The pipeline depends on unified perception methods which may be disrupted.
- In certain conditions, objects may not be accurately recognized, leading to failures.
- Physical dynamics can result in objects being dropped during the grasping process.

Application in Motion



(a video of the fetch robot in action)

Visit Our Website!

https://tinyurl.com/4zk277bk

SCAN ME

Citations

- [1] Bowen Wen, Wei Yang, Jan Kautz, Stan Birchfield et al., “Foundationpose: Unified 6d pose estimation and tracking of novel objects,” in CVPR, 2024.
- [2] Yangxiao Lu, Yunhui Guo, Nicholas Ruozzi, Yu Xiang, et al. “Adapting pre-trained vision models for novel instance detection and segmentation,” arXiv preprint arXiv:2405.17859, 2024.
- [3] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo et al., “Segment anything,” in ICCV, 2023.

THANK YOU!

CREDITS: This presentation template was created by Slidesgo, including icons by Flaticon, infographics & images by Freepik